

Election

Tags:

Time Limit: 1 s Memory Limit: 128 MB
Submission: 25 AC: 15 Score: 99.85

Submit

Codes

AI Code Tips

Description

Problem C - Election

Canada has a multi-party system of government. Each candidate is generally associated with a party, and the party whose candidates win the most ridings generally forms the government. Some candidates run as independents, meaning they are not associated with any party. Your job is to count the votes for a particular riding and to determine the party with which the winning candidate is associated.

Input

The first line of input contains a positive integer n satisfying $2 \leq n \leq 20$, the number of candidates in the riding. n pairs of lines follow: the first line in each pair is the name of the candidate, up to 80 characters; the second line is the name of the party, up to 80 characters, or the word "independent" if the candidate has no party. No candidate name is repeated and no party name is repeated in the input. No lines contain leading or trailing blanks.

Output

Output consists of a single line containing one of:

- The name of the party with whom the winning candidate is associated, if there is a winning candidate and that candidate is associated with a party.
- The word "independent" if there is a winning candidate and that candidate is not associated with a party.
- The word "tie" if there is no winner; that is, if no candidate receives more votes than every other candidate.

Samples

input

Copy

```
3
Marilyn Manson
Rhinoceros
Jane Doe
Family Coalition
John Smith
independent
6
John Smith
```

```
Marilyn Manson
Marilyn Manson
Jane Doe
John Smith
Marilyn Manson
```

output Copy

```
Rhinoceros
```

Submit

Codes

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Server Time: 2025-8-26 15:10:46